Software Testing Lab 7

Fuzz Testing

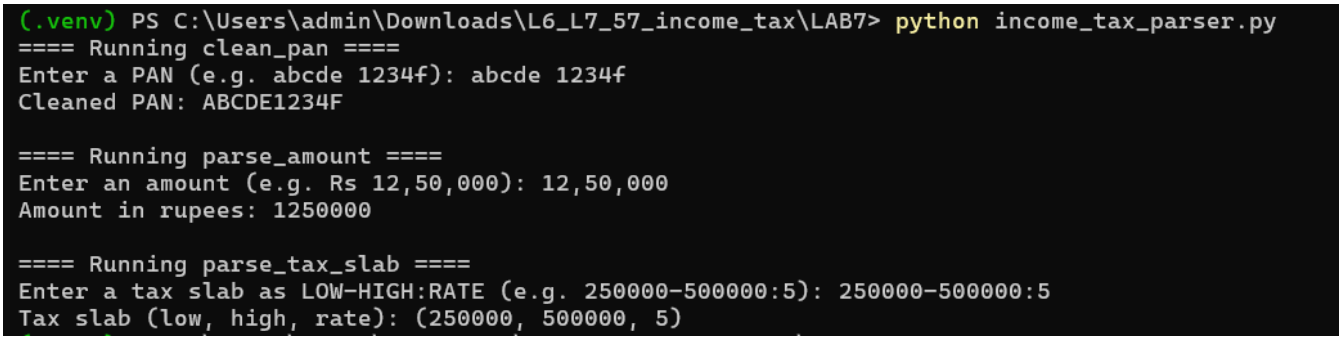
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Income tax input validation with Hypothesis

Test Evidence

SS1 - Original application



Test Evidence

SS2 - Hypothesis failures

```
(.venv) PS C:\Users\admin\Downloads\L6_L7_57_income_tax\LAB7> python -m pytest test_income_tax_parser.py -v
============================= test session starts ==============================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\admin\Downloads\L6_L7_57_income_tax\LAB6\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\admin\Downloads\L6_L7_57_income_tax\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_income_tax_parser.py::test_clean_pan_no_crash FAILED                [ 33%]
test_income_tax_parser.py::test_parse_amount_no_crash FAILED             [ 66%]
test_income_tax_parser.py::test_parse_tax_slab_no_crash FAILED           [100%]

================================== FAILURES ===================================
_____________________________ test_clean_pan_no_crash _____________________________

s = None

    @given(st.text() | st.none())
    def test_clean_pan_no_crash(s):
        try:
>           clean_pan(s)

test_income_tax_parser.py:8:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

pan = None

    def clean_pan(pan):
        """
        Removes spaces from a PAN and upper-cases it.
        Assumes pan is a non-empty string.
        """
>       return pan.strip().replace(' ', '').upper()
               ^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

income_tax_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_clean_pan_no_crash(s):
                ^^^

test_income_tax_parser.py:6:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @given(st.text() | st.none())
    def test_clean_pan_no_crash(s):
        try:
            clean_pan(s)
        except Exception as e:
>           assert False, f"clean_pan crashed with input {s!r}: {e}"
E           AssertionError: clean_pan crashed with input None: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_clean_pan_no_crash(
E               s=None,
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   C:\Users\admin\Downloads\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py:9
```

Test Evidence

SS3 - Fixed application

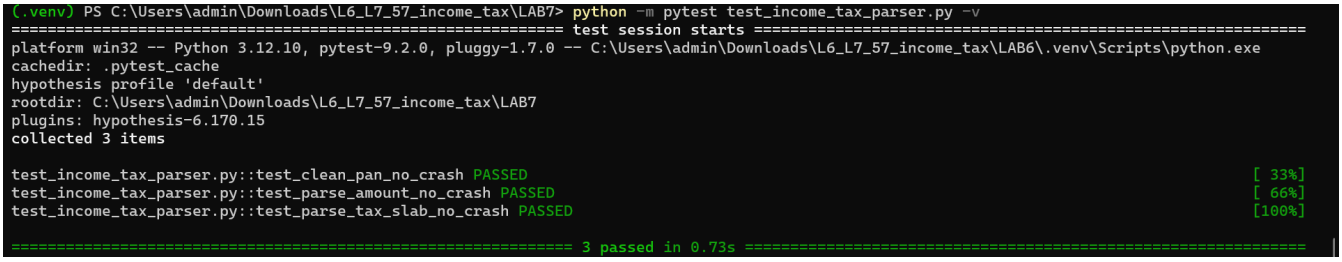
```
(.venv) PS C:\Users\admin\Downloads\L6_L7_57_income_tax\LAB7> python income_tax_parser.py
==== Running clean_pan ====
Enter a PAN (e.g. abcde 1234f): abcde 1234f
Cleaned PAN: ABCDE1234F

==== Running parse_amount ====
Enter an amount (e.g. Rs 12,50,000): 12,50,000
Amount in rupees: 1250000

==== Running parse_tax_slab ====
Enter a tax slab as LOW-HIGH:RATE (e.g. 250000-500000:5): 250000-500000:5
Tax slab (low, high, rate): (250000, 500000, 5)
```

Test Evidence

SS4 - Passing tests



Reflection

1. What is fuzz testing, and why is it useful?

Fuzz testing supplies random, unexpected, or malformed inputs to expose crashes and edge cases that manual tests may miss.

2. How did Hypothesis help identify bugs?

Hypothesis generated values such as None, empty strings, and malformed tax slab inputs. It exposed failures in all three functions and minimized failing examples.

3. How did you improve the program's robustness?

Type checks, exception handling, and format validation were added. Invalid inputs return safe defaults, and all three Hypothesis tests pass.